

Zirconium

Zr (Zr I). Strongest classified transitions below 2 μm (NIST): 40 levels, 0 lines.

Zr⁺ ionisation limit 53507.83 cm⁻¹ = 6.63413 eV (186.889 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| e r \| J' \rangle |$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).

